How do we know what we know about populations?

Census Data, Vital Records, Immigration

Lecture 1.3, Jan 27th, 2026

Motivation:
Who cares how we measure populations?

- Anyone interested in allocation of resources
  - Businesses looking to invest (should I build an apartment building here? Open a grocery store there?)
  - Governments thinking about the future (School here? Hospital there? How many old people need SSA?)
  - Investors thinking about a stock market portfolio (which demographics are growing, and which companies sell things to them?)

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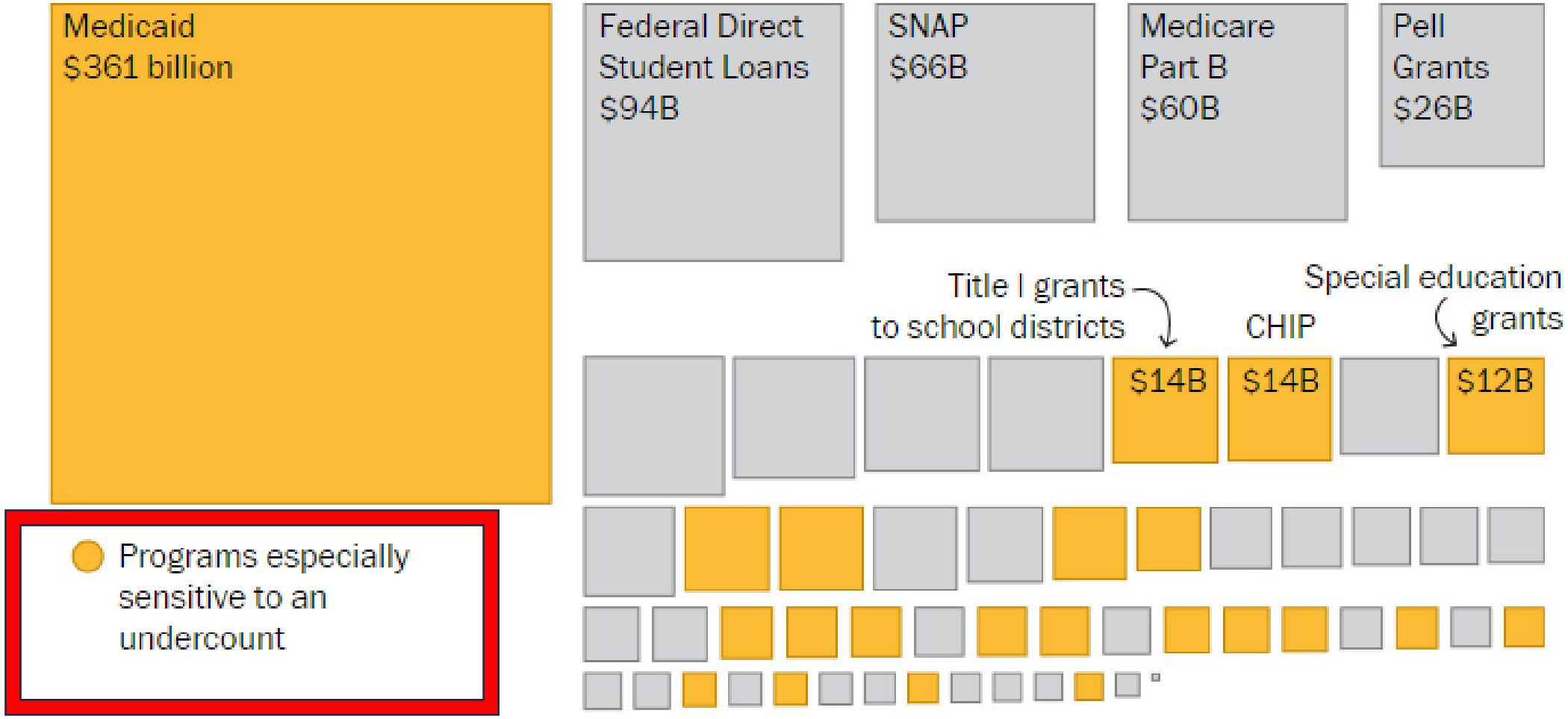
- Data are inherently *political*
- Data have *political uses*

What happens when we undercount?

55 large census-reliant federal programs distributed $883 billion in 2016



So how do we count?

Recall: Growth Balance Equation

Natural Increase

$$\text{Population}_{T2} = \text{Population}_{T1} + \boxed{B - D} + \boxed{I - O}$$

Net Migration

Each piece of the equation has to be counted

Where does population data come from?

$$\boxed{P_2 = P_1} + B - D + I - O$$

- Census
  - Full-count census every 10 years, mandated in the constitution

US Constitution Article 1, Section 2 amended by section 2 of the 14th amendment

"Representatives shall be apportioned among the several States according to their representative numbers, counting the whole number of persons in each State, excluding Indians not taxed"

Two big reasons this is in the constitution:

1) Allocation of seats in for states House of Representatives
2) Taxation + spending

Where does population data come from?

$$\boxed{P_2 = P_1} + B - D + I - O$$

- Census
  - Full-count census every 10 years, mandated in the constitution
  - Other countries have something like it
  - Supplemented by more frequent surveys

Where does population data come from?

$$\boxed{P_2 = P_1} + \boxed{B - D} + I - O$$

- Census
- Vital registration system (birth certificate, death certificate)

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- Indirect techniques
  - Surveys
  - Interpolation

Where does population data come from?

$$\boxed{P_2 = P_1} + \boxed{B - D} + \boxed{I - O} + X$$

- Census
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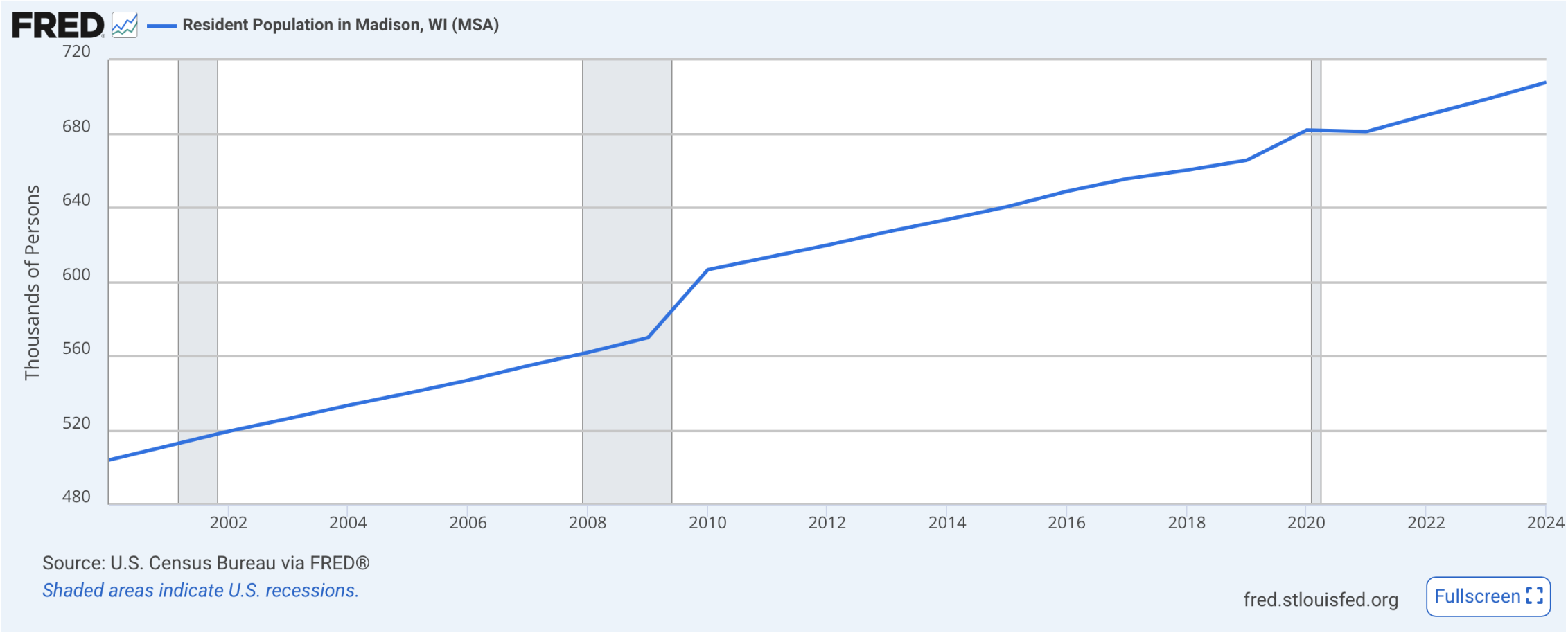
Error of closure

When a new Census is conducted, errors become apparent (where people were not counted)



Data can be incomplete and messy, especially historical population data

Ex: A fire in January 1921 destroyed almost all records of the 1890 census

Data quality depends on the available infrastructure

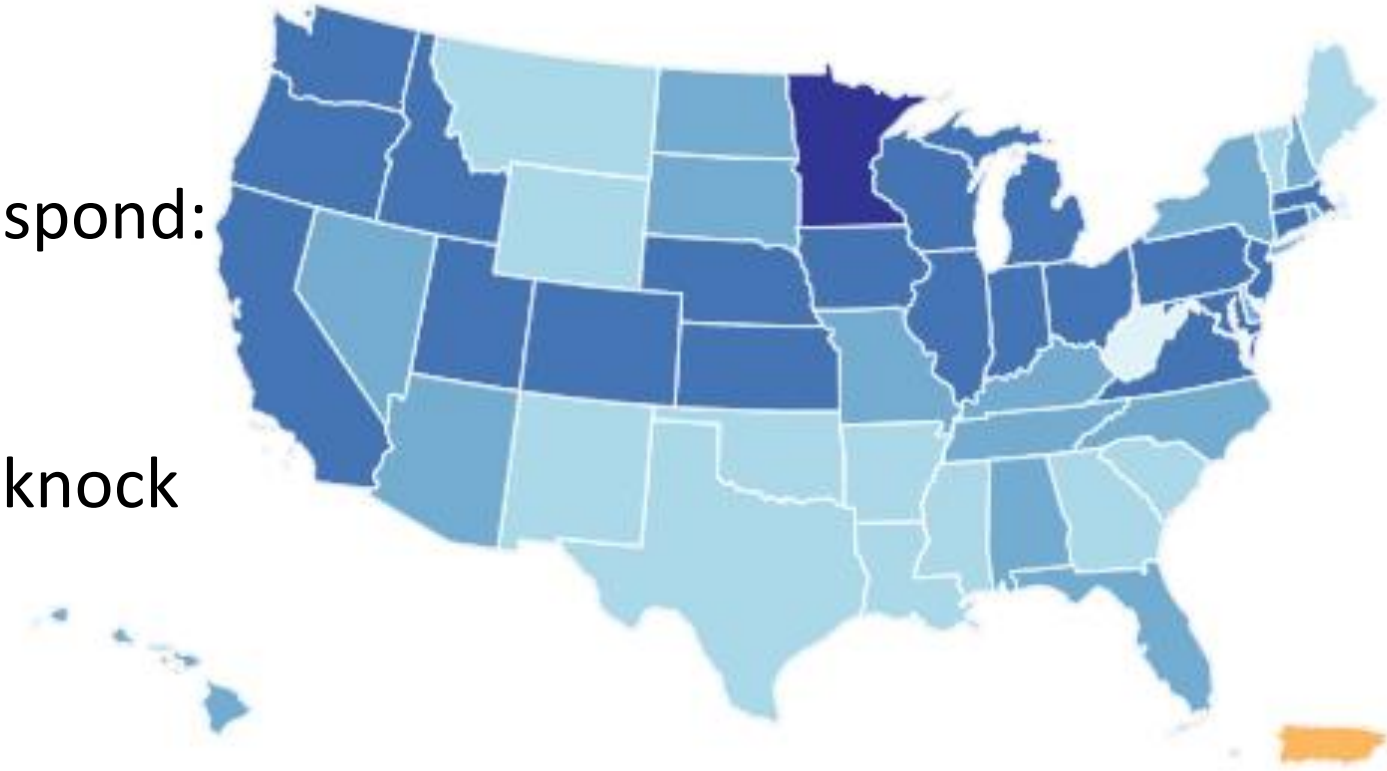
How reliable is data collection in the US Census?

- Response rates and census hesitancy
- What questions are asked - or not asked - and how are they asked?
  - Differential Undercounts
- Changes to the Census over time

2020 Census Self-Response Rate

If an address doesn't respond:

1) Try again (mail)
2) Try again (mail)
3) Send someone to go knock
4) Knock again
5) Knock again
6) Knock again
7) Knock again
8) Knock again



Self-Response Rate (%)

0-15 | 16-30 | 31-40 | 41-50 | 51-56 | 57-62 | 63-68 | 69-74 | 75-85 | 86-100

Differential undercounts: Hesitancy? Structural barriers to counts?

- Some sub-populations are more likely to be undercounted than others

Undercount rates still VASTLY higher for Black Americans

Table 2. Percentage Net Decennial Census Undercount by Race, as Estimated by Demographic Analysis, 1940 through 2000

	1940	1950	1960	1970	1980	1990	2000
Total population	5.4%	4.1%	3.1%	2.7%	1.2%	1.65%	0.12%
Black	8.4%	7.5%	6.6%	6.5%	4.5%	5.52%	2.78%
Non-Black	5.0%	3.8%	2.7%	2.2%	0.8%	1.08%	-0.29%

Sources: Estimates for 1940 through1980 are from J.G. Robinson, et al., "Estimates of Population Coverage in the 1990 United States Census Based on Demographic Analysis," *Journal of the American Statistical Association*, vol. 88 (September 1993), p. 1065, reprinted in U.S. Bureau of the Census, *Accuracy and Coverage Evaluation, Statement on the Feasibility of Using Statistical Methods to Improve the Accuracy of Census 2000*, June 2000 (unpublished document). Estimates for 1990 and 2000 are from U.S. Bureau of the Census, *Coverage Measurement from the Perspective of March 2001 Accuracy and Coverage Evaluation*, Census 2000 Topic Report no. 4 (Washington: U.S. Bureau of the Census, February 2004), p. 9.

Undercount discovered when men had to sign up for the draft.

Census counted too few Black and Hispanic people in the US

The census has trouble accurately counting people of some races and ethnicities. In 2020, the official census counted **0.2% fewer people overall** in the U.S. than are actually in the country. And it counted 1.6% more **non-Hispanic whites** than there actually are, and 5.6% fewer **American Indian and Alaskan Natives on reservations** than there are.

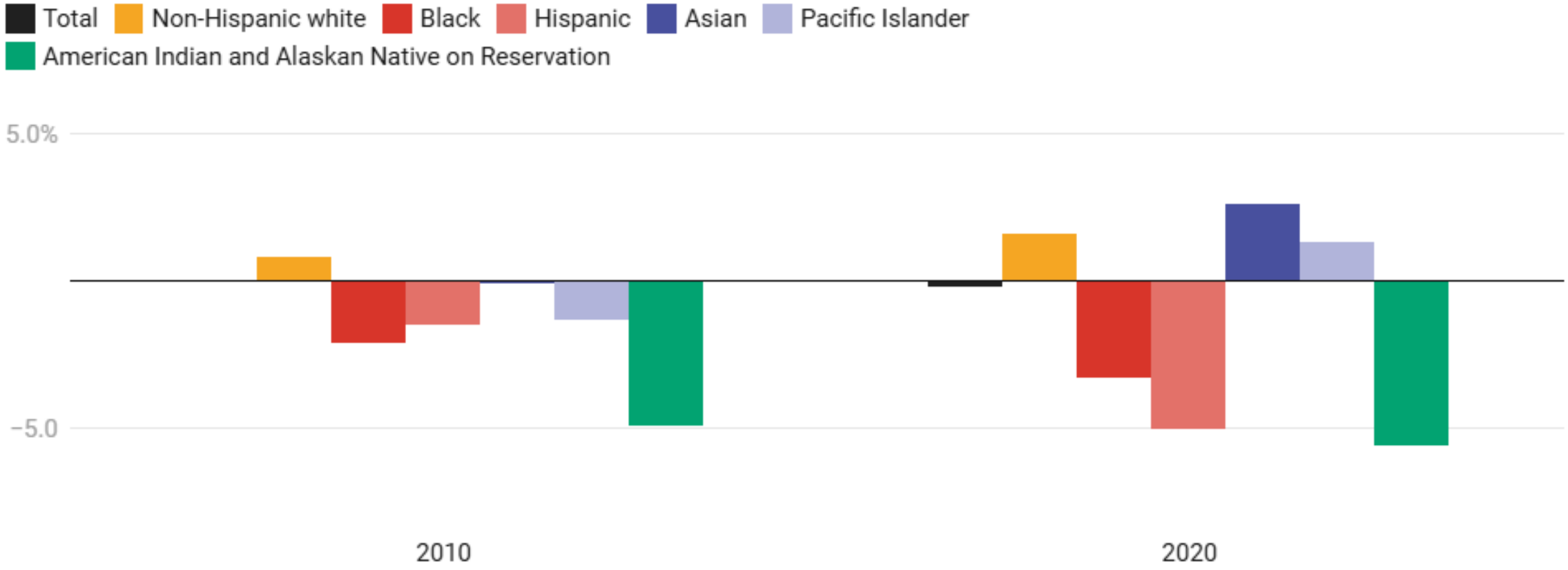


Chart: The Conversation, CC-BY-ND • Source: U.S. Census Bureau • Get the data • Download image

TABLE 2. PES PERCENT NET COVERAGE ERROR BY RACE AND HISPANIC ORIGIN: 2010 AND 2020

	2010	2020	2020 Significantly Different from 2010
Total	**0.01**	**-0.24**	
Race Alone or in Combination			
White	0.54*	0.66*	No
Non-Hispanic White alone	0.83*	1.64*	Yes
Black or African American	-2.06*	-3.30*	No
Asian	0.00	2.62*	Yes
American Indian or Alaska Native	-0.15	-0.91*	No
On Reservation	-4.88*	-5.64*	No
American Indian Areas Off Reservation	3.86	3.06	No
Balance of the United States	0.05	-0.86*	No
Native Hawaiian or Other Pacific Islander	-1.02	1.28	No
Some Other Race	-1.63*	-4.34*	Yes
Hispanic or Latino	-1.54*	-4.99*	Yes

Reasons for under/overcount:

1) Babies
2) Elderly
3) People move
4) Children w/ multiple households
5) Response Hesitancy (system avoidance)

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These are BIG numbers!
Hispanic population ~60 million
So 5% undercount is ~3 million people!

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How reliable is data collection in the US Census?

- Response rates and census hesitancy
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  - Differential Undercounts
- How are questions asked?

Example: Citizenship Question

Decision to add citizenship question to census draws protest

By KEVIN FREKING and HOPE YEN March 27, 2018

APR. 23, 2019, AT 9:38 AM

How The Citizenship Question Could Break The Census

By Amelia Thomson-DeVeaux

Filed under Supreme Court

Why census experts fear a citizenship question would jeopardize results

Apr 23, 2019 6:45 PM EST

Supreme Court rules against Trump on census citizenship question

BY JACQUELINE THOMSEN - 06/27/19 10:44 AM EDT

14,645 COMMENTS

Census citizenship question could create fear for Madison immigrants, lead to undercount

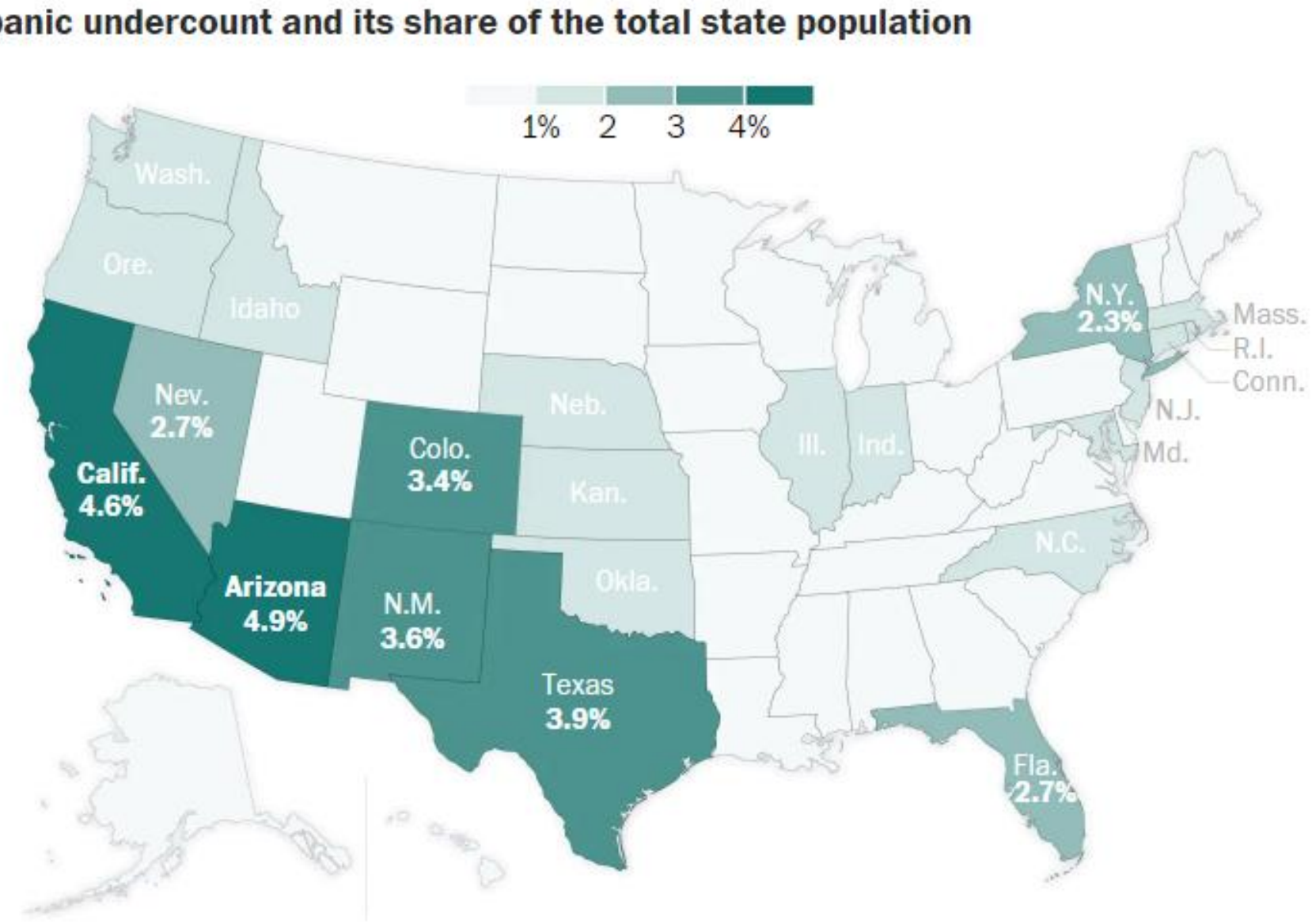
Abigail Becker | The Capital Times Jun 25, 2019 6 min to read

Example: Citizenship Question

- Fears it would lead to:
  - People refusing to fill out the form
  - Misreporting foreign-born status
- US already knows citizenship of about 90% US residents
  - Estimates say that the question would have only bumped that to about 93%.

Potential effects of citizenship question:

Has implications for representation in Congress, funding, etc



[Where a citizenship question could cause the census to miss millions of Hispanics - The Washington Post]

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Hispanic undercount increased 2.5x!

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Limitations to how questions are asked

6. What is Person 1's sex? *Mark* ✗ *ONE box.*

☐ Male ☐ Female

- Best estimates are that 1.7% of people are born with intersex traits
  - ~5 million Americans!
- .5% clinically identifiable sexual or reproductive variations
  - ~1.7 million Americans
- ~1.3 million American adults identify as transgender

2010: Had to infer same-sex relationships

2020: Same-sex relationships explicit

2. How is this person related to Person 1? *Mark* ✗ *ONE box.*

- ☐ Husband or wife
- ☐ Biological son or daughter
- ☐ Adopted son or daughter
- ☐ Stepson or stepdaughter
- ☐ Brother or sister
- ☐ Father or mother
- ☐ Grandchild
- ☐ Parent-in-law
- ☐ Son-in-law or daughter-in-law
- ☐ Other relative
- ☐ Roomer or boarder
- ☐ Housemate or roommate
- ☐ Unmarried partner
- ☐ Other nonrelative

Source: U.S. Census Bureau, 2010 Census questionnaire.

How is this person related to Person 1? *Mark* ✗ *ONE box.*

- ☐ Opposite-sex husband/wife/spouse
- ☐ Opposite-sex unmarried partner
- ☐ Same-sex husband/wife/spouse
- ☐ Same-sex unmarried partner
- ☐ Biological son or daughter
- ☐ Adopted son or daughter
- ☐ Stepson or stepdaughter
- ☐ Brother or sister
- ☐ Father or mother
- ☐ Grandchild
- ☐ Parent-in-law
- ☐ Son-in-law or daughter-in-law
- ☐ Other relative
- ☐ Roommate or housemate
- ☐ Foster child
- ☐ Other nonrelative

Source: US Census Bureau, 2020 Census questionnaire.

Changing Measurement of Coupled Households

1880 Census

- Relationship to the householder first appears on the Census:

  Relationship of each person to the head of this family – whether wife, son, daughter, servant, boarder, or other

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1990 Census

- Added the category unmarried partner
- If reported as same-sex:
  - Unmarried partners allowed
  - Married couples:

    Sex of one individual edited to reflect opposite-sex arrangement

NEW in 1990:
Person of Opposite Sex Sharing Living Quarters

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2010 Census

- First to show estimates of same-sex married couples in special report/tables
- Revised estimates released to address an error in the estimates

Similar Issues With Vital Statistics:

Scenario I

CAUSE OF DEATH (See instructions and examples)

32. PART I. Enter the chain of events--diseases, injuries, or complications--that directly caused the death. DO NOT enter terminal events such as cardiac arrest, respiratory arrest, or ventricular fibrillation without showing the etiology. DO NOT ABBREVIATE. Enter only one cause on a line. Add additional lines if necessary.

		Approximate Interval: Onset to death
IMMEDIATE CAUSE (Final disease or condition --------> resulting in death)	a. Acute respiratory acidosis	3 days
	Due to (or as a consequence of):	
Sequentially list conditions, if any, leading to the cause listed on line a. Enter the **UNDERLYING CAUSE** (disease or injury that initiated the events resulting in death) **LAST**	b. COVID-19	1 week
	Due to (or as a consequence of):	
	c.	
	Due to (or as a consequence of):	
	d.	

PART II. Enter other significant conditions contributing to death but not resulting in the underlying cause given in PART I

Chronic obstructive pulmonary disease, hypertension

33. WAS AN AUTOPSY PERFORMED? ☐ Yes ■ No

34. WERE AUTOPSY FINDINGS AVAILABLE TO COMPLETE THE CAUSE OF DEATH? ☐ Yes ☐ No

35. DID TOBACCO USE CONTRIBUTE TO DEATH?

☐ Yes ☐ Probably

■ No ☐ Unknown

36. IF FEMALE:

☐ Not pregnant within past year

☐ Pregnant at time of death

☐ Not pregnant, but pregnant within 42 days of death

☐ Not pregnant, but pregnant 43 days to 1 year before death

☐ Unknown if pregnant within the past year

37. MANNER OF DEATH

■ Natural ☐ Homicide

☐ Accident ☐ Pending Investigation

☐ Suicide ☐ Could not be determined

Table. Number of Deaths for Leading Causes of Death, US, 2015-2020[a]

Big jumps in these too!

	No. of deaths by year					
Cause of death	2015	2016	2017	2018	2019	2020
Total deaths	2 712 630	2 744 248	2 813 503	2 839 205	2 854 838	3 358 814
Heart disease	633 842	635 260	647 457	655 381	659 041	690 882
Cancer	595 930	598 038	599 108	599 274	599 601	598 932
COVID-19[b]						345 323
Unintentional injuries	146 571	161 374	169 936	167 127	173 040	192 176
Stroke	140 323	142 142	146 383	147 810	150 005	159 050
Chronic lower respiratory diseases	155 041	154 596	160 201	159 486	156 979	151 637
Alzheimer disease	110 561	116 103	121 404	122 019	121 499	133 382
Diabetes	79 535	80 058	83 564	84 946	87 647	101 106
Influenza and pneumonia	57 062	51 537	55 672	59 120	49 783	53 495
Kidney disease	49 959	50 046	50 633	51 386	51 565	52 260
Suicide	44 193	44 965	47 173	48 344	47 511	44 834

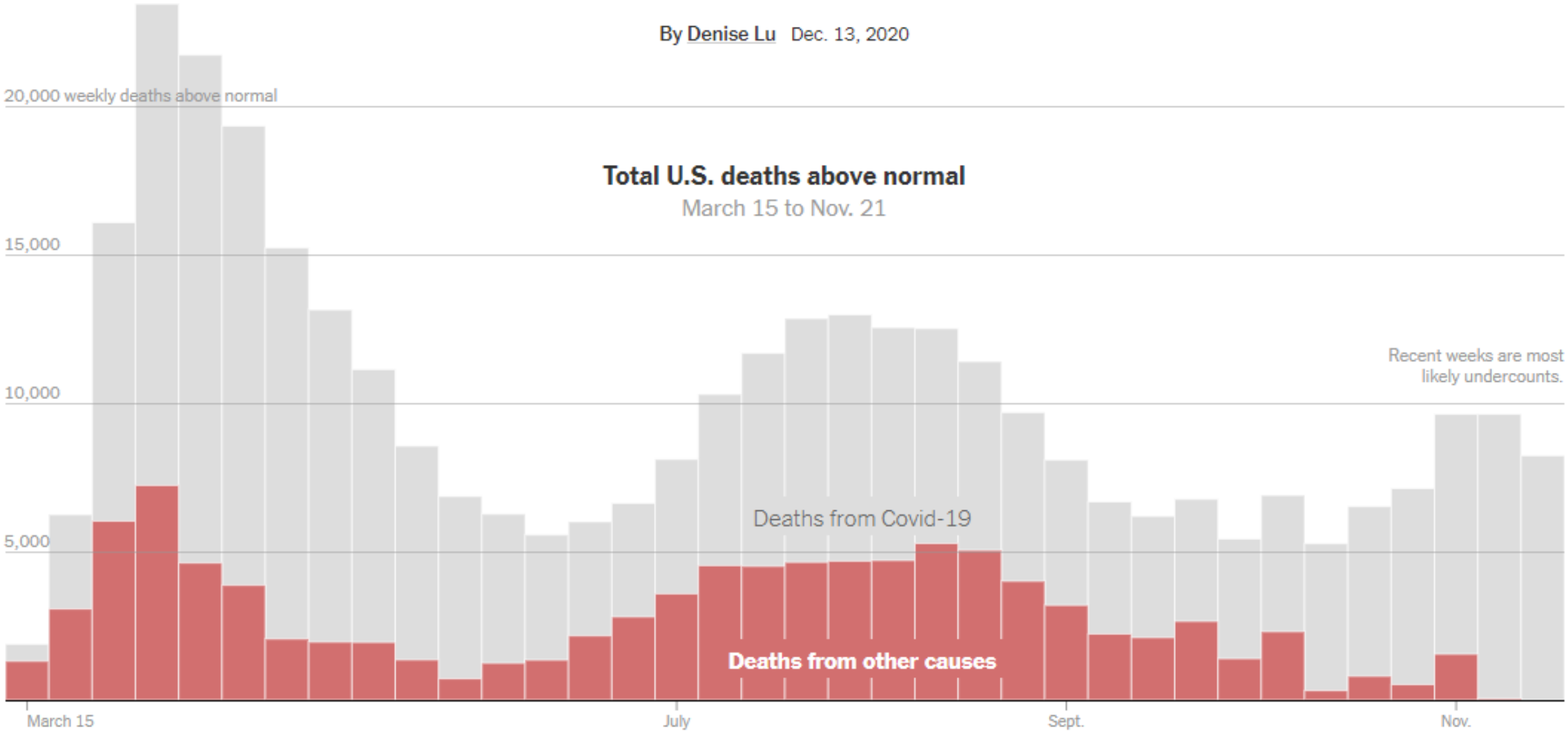
[a] Leading causes are classified according to underlying cause and presented according to the number of deaths among US residents. For more information, see the article by Heron.[4] Source: National Center for Health Statistics. National Vital Statistics System: mortality statistics (http://www.cdc.gov/nchs/deaths.htm). Data for 2015-2019 are final; data for 2020 are provisional.

[b] Deaths with confirmed or presumed COVID-19, coded to *International Statistical Classification of Diseases and Related Health Problems, Tenth Revision* code U07.1 as the underlying cause of death.

Source: Ahmad and Anderson. 2021. *JAMA* 325(18): 1829-1830

2020 Was Especially Deadly. Covid Wasn't the Only Culprit.

By Denise Lu Dec. 13, 2020



Was COVID-19 the Leading Cause of Death in 2020?

Cause of Death (*ICD-10* Code)	No. of Deaths
National Center for Health Statistics list[a]	
Heart disease (I00–I09, I11, I13, I20–I51)	690 882
Cancer (C00–C97)	598 932
COVID-19 (U07.1)	435 323
Unintentional injuries (V01–X59, Y85–Y86)	192 176
Chronic lower respiratory diseases (J40–J47)	159 050
Stroke (I60–I69)	151 637
Alzheimer's disease (G30)	133 382
Diabetes (E10–E14)	101 106
Influenza and pneumonia (J09–J18)	53 495
Kidney disease (N00–N07, N17–N19, N25–N27)	52 260
Suicide (U03, X60–X84, Y870)	44 834

World Health Organization list[b]	
COVID-19 (U07.1)	435 323
Ischemic heart diseases (I20–I25)	369 579
Dementia and Alzheimer's disease (F01, F03, G30)	242 273
Chronic lower respiratory diseases (J40–J47)	159 050
Stroke (I60–I69)	151 637
Lung cancer (C33-C34)	143 041
Heart failure (I50–I51)	108 866
Hypertensive diseases (I10–I15)	104 527
Diabetes (E10–E14)	101 106
Diseases of the urinary system (N00–N39)	73 037
Accidental poisoning (X40–X49)	67 355

Source: Tai and Lu 2021. AJPH

Estimating Rates: Combining Sources

$$\text{General fertility rate} = \frac{\text{\# births in year}}{\text{\# women aged 15-49 mid-year}}$$

Requires data from vital registration (#births)

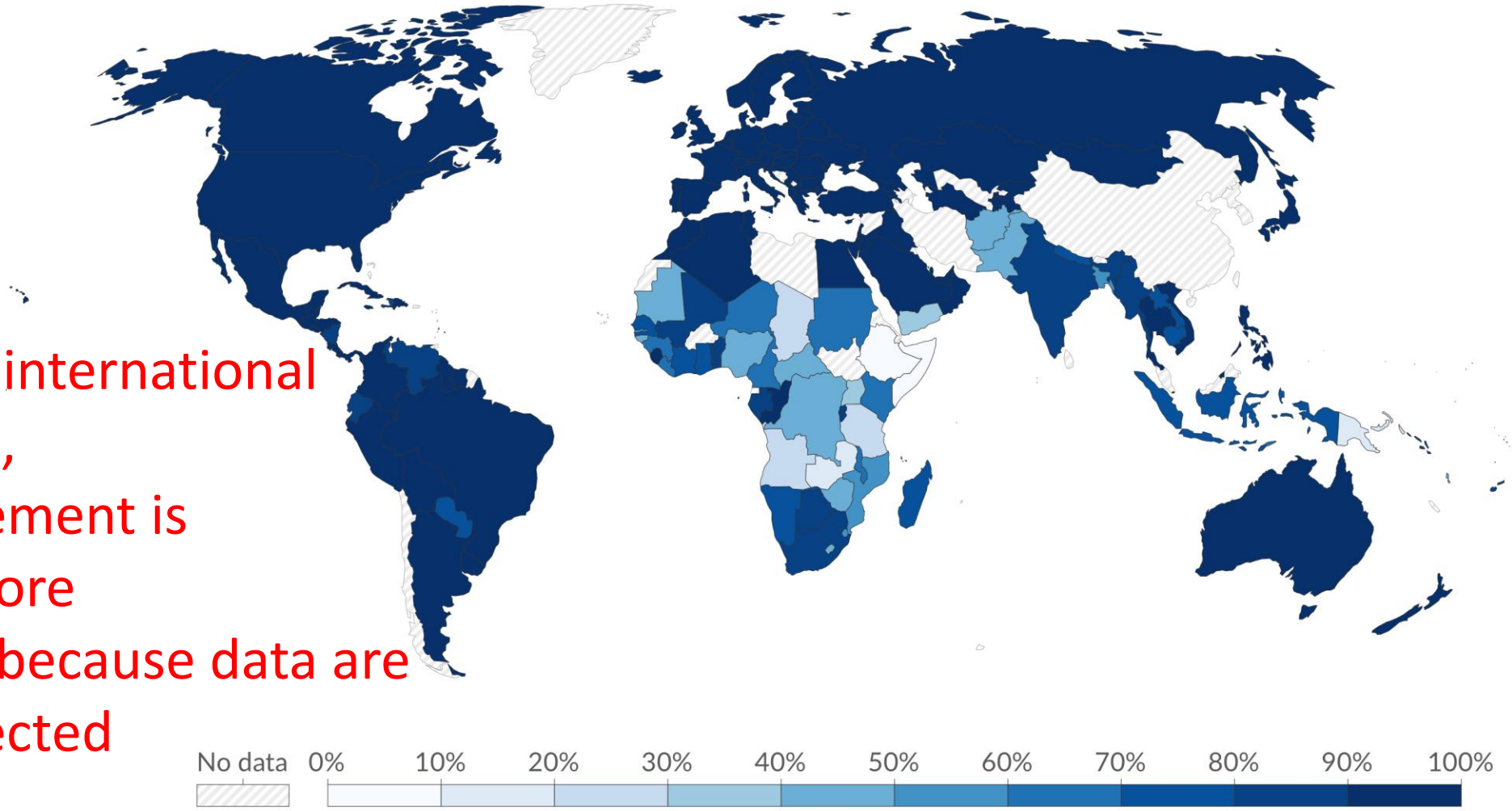
AND from Census (#population)

Important to think about limitations of BOTH data sources!

Share of births registered, 2022

Our World in Data

Percentage of children under age five whose births were registered at the time of the survey. This includes children whose birth certificate was seen by the interviewer or whose mother or caretaker says the birth has been registered.



In some international contexts, measurement is much more difficult because data are not collected

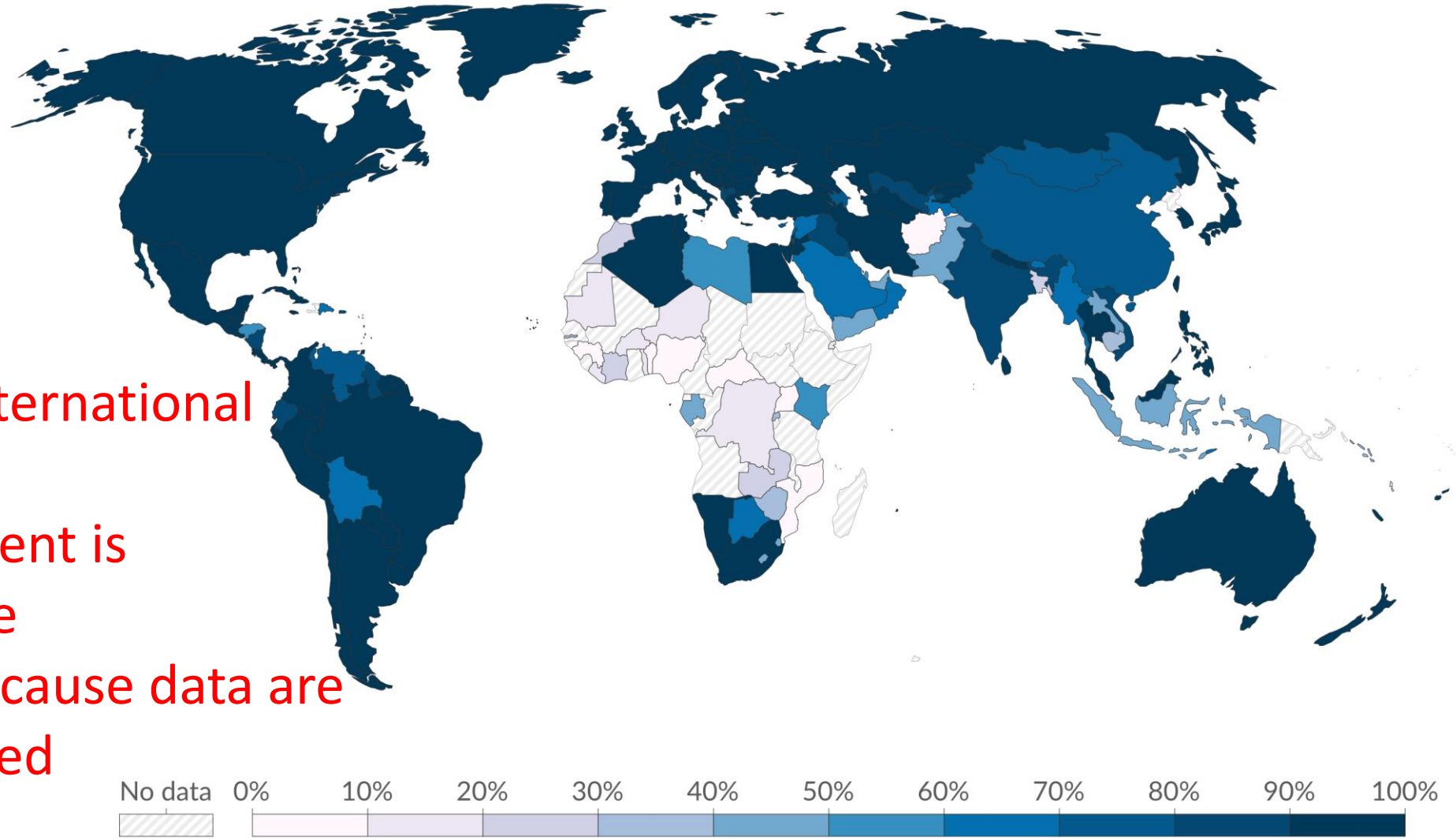
Data source: Household surveys (DHS, MICS), compiled by UNICEF, via World Bank (2025)

OurWorldinData.org/population-growth | CC BY

Share of deaths that are registered, 2019

Deaths reported in a country's vital registration system[1] as a share of total expected deaths.

Our World in Data

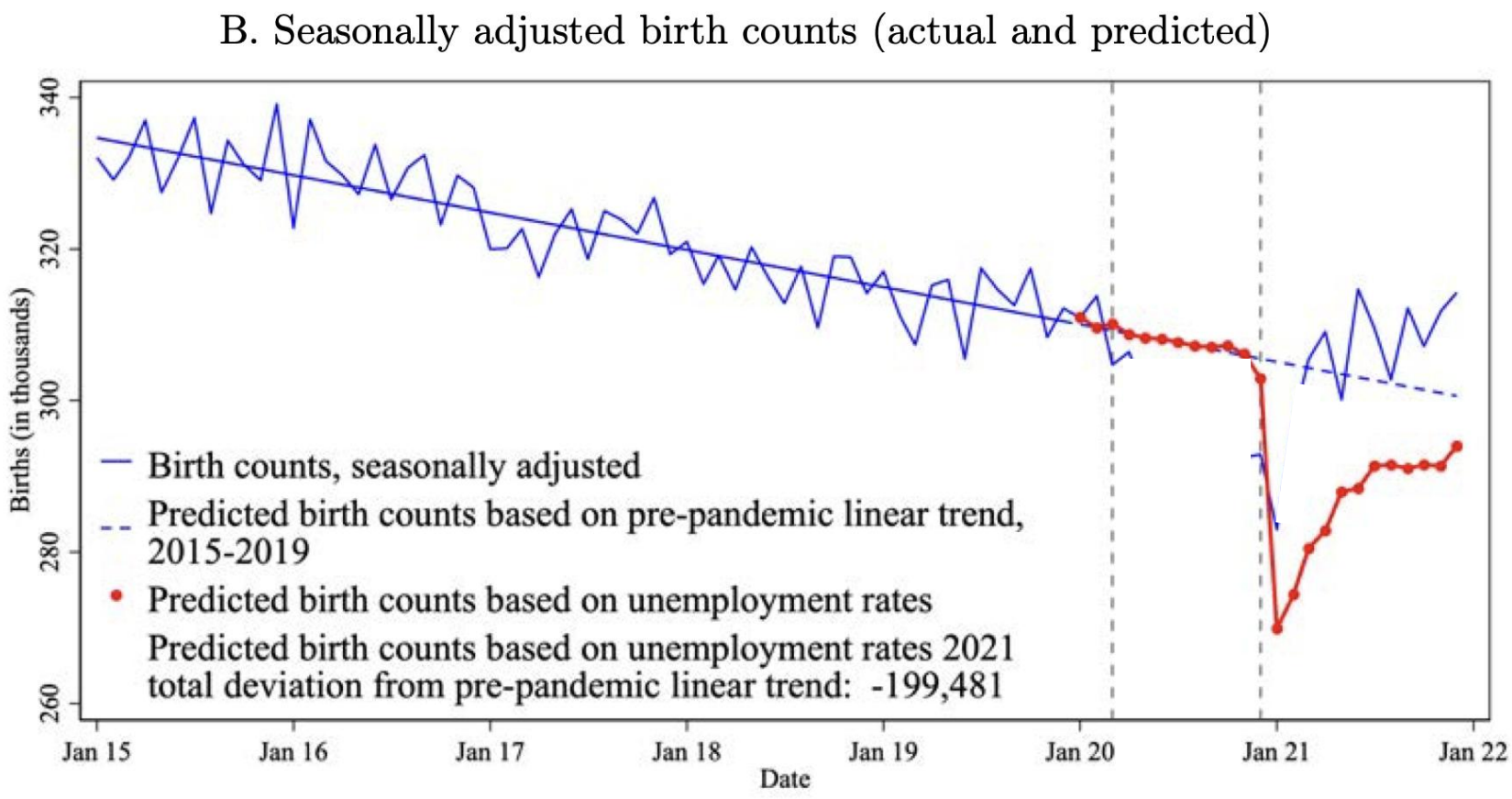


In some international contexts, measurement is much more difficult because data are not collected

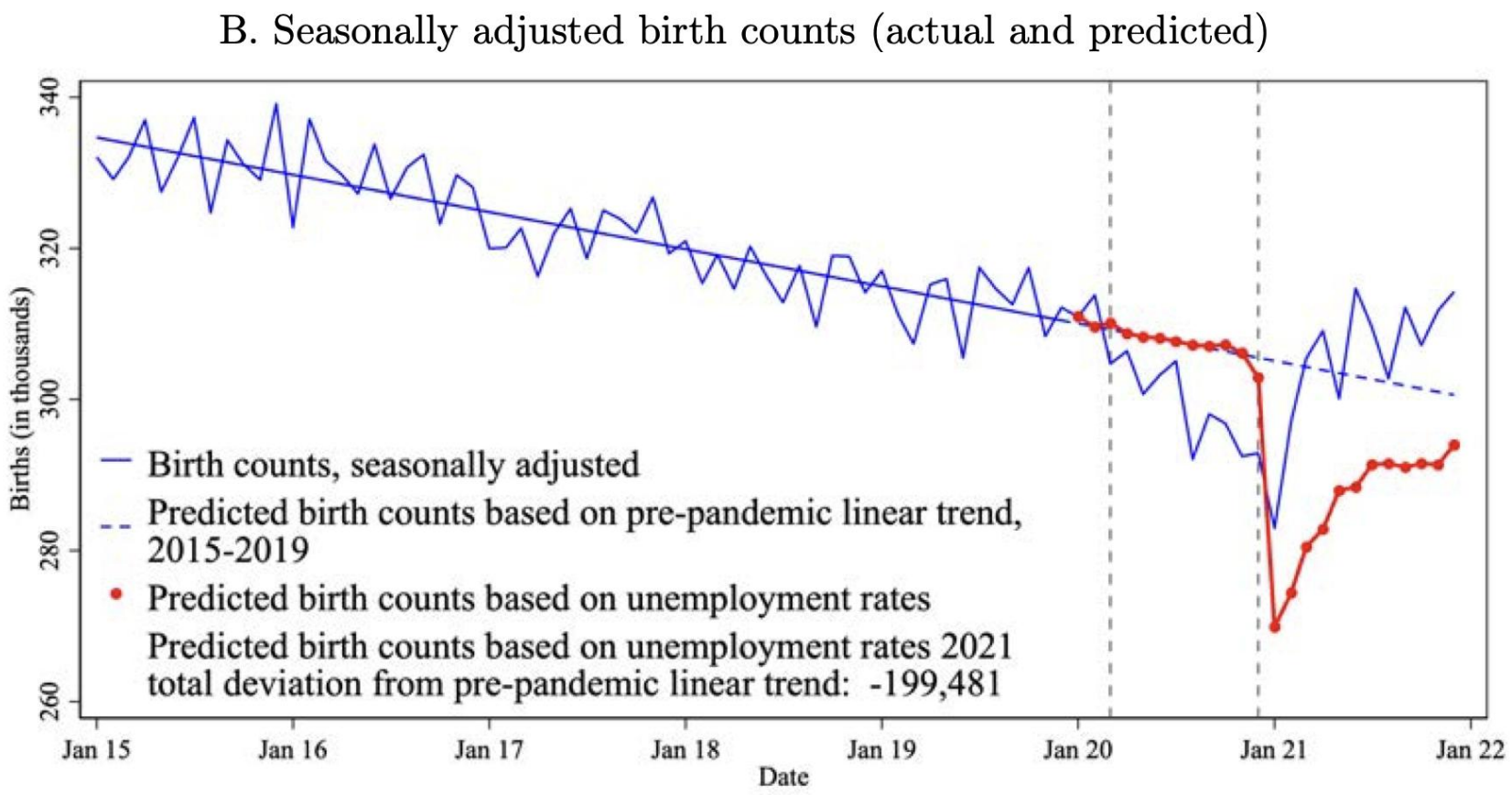
Data source: Ariel Karlinksy (2024)

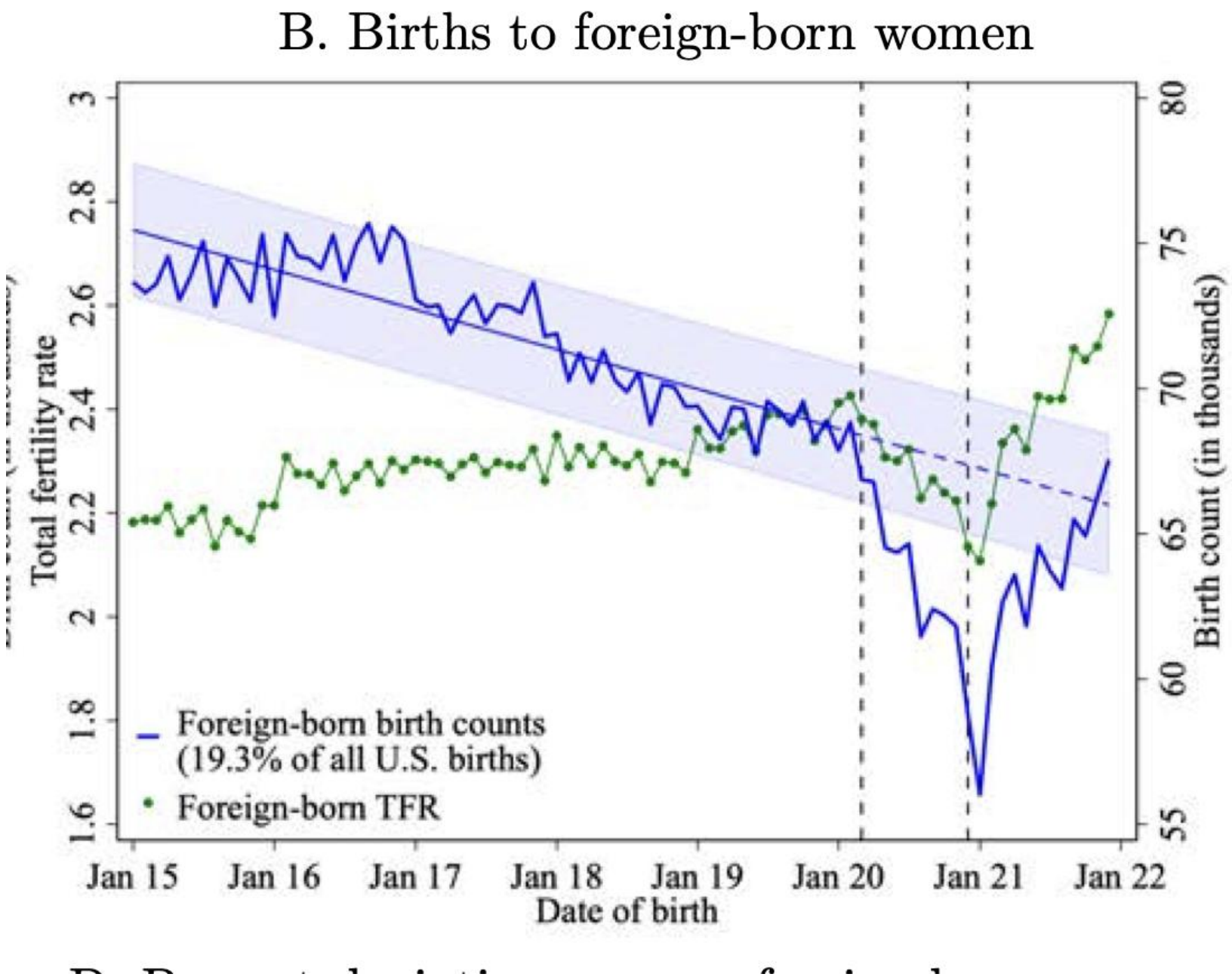
OurWorldinData.org/causes-of-death | CC BY

An Example Where Data Mattered: Births During Covid



Decline in births began right away, not 9 months after Covid!





Lots of foreign-born people either:

A) left the US in anticipation of a lockdown or

B) didn't come to the US because of the lockdown

Other Data Collection Efforts in the US (I will not test you on these!)

- The census department runs >90 surveys across many topics
  - American Community Survey (ACS) is an annual version of the Census w/ extra info
  - Current Population Survey (CPS) is a monthly version conducted by the Census for the Bureau of Labor Statistics, with supplements that cover specific topics like employment
    - This is where unemployment stats come from
    - Businesses, investors use these data for planning
  - Economic Census (EC) is a census of businesses in the US conducted every five years

Questions of the day

https://tinyurl.com/jan27soc170

Why is it important for the government to collect accurate and timely data? (1 sentence)

Name two primary sources of data used to calculate demographic rates.